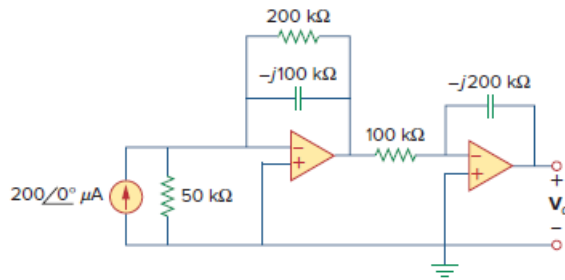


Problem

For the op amp circuit in Fig. 10.122, obtain V_o .

Figure 10.122:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.122 in the textbook.

First convert the current source into voltage source using source transformation.

$$\begin{aligned} V_s &= (200 \times 10^{-6} \angle 0^\circ)(50 \times 10^3) \\ &= 10 \angle 0^\circ \text{ V} \end{aligned}$$

Draw the modified circuit.

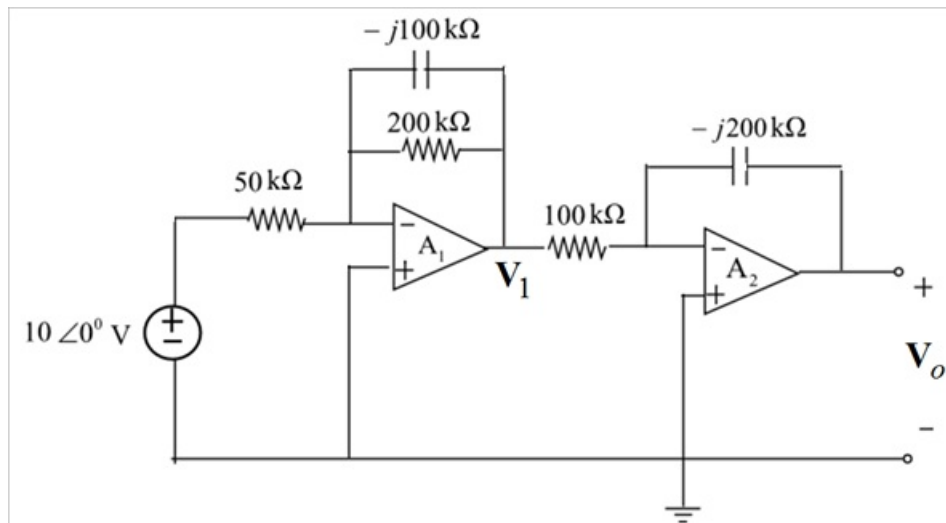


Figure 1

Step 2 of 4

Amplifier A_1 in Figure 1 is an inverting amplifier.

Thus, the output of the op-amp A_1 is,

$$V_1 = -\frac{Z_2}{Z_1} V_s \dots\dots (1)$$

Calculate the feedback impedance of amplifier 1.

$$\begin{aligned} Z_2 &= (200 \text{ k}\Omega) \parallel (-j100 \text{ k}\Omega) \\ &= \frac{(200 \times 10^3) \times (-j100 \times 10^3)}{(200 \times 10^3) + (-j100 \times 10^3)} \\ &= 89443 \angle -63.43^\circ \Omega \end{aligned}$$

Calculate the input impedance of amplifier 1.

$$Z_1 = 50 \text{ k}\Omega$$

Step 3 of 4

From equation (1),

$$\begin{aligned} V_1 &= -\frac{89443 \angle -63.43^\circ}{50000} (10 \angle 0^\circ) \\ &= 17.89 \angle 116.57^\circ \text{ V} \end{aligned}$$

The amplifier A_2 in figure 1 is an inverting amplifier.

Thus, the output of the op-amp A_2 is,

$$V_o = -\frac{Z_4}{Z_3} V_1 \dots\dots (2)$$

From Figure 1, the feedback impedance of amplifier 2 is,

$$Z_4 = -j200 \text{ k}\Omega$$

The input impedance of amplifier 2 is,

$$Z_3 = 100 \text{ k}\Omega$$

Step 4 of 4

From equation (2), the output voltage is,

$$\begin{aligned} V_o &= -\left(\frac{-j200 \times 10^3}{100 \times 10^3}\right) (17.89 \angle 116.57^\circ) \\ &= (2 \angle 90^\circ) (17.89 \angle 116.57^\circ) \\ &= 35.8 \angle -153.44^\circ \text{ V} \end{aligned}$$

Therefore, the output voltage, V_o is $\boxed{35.8 \angle -153.44^\circ \text{ V}}$.